

Generation Interconnection System Impact Study Report

For Queue Project Q12-0134

Podick 100 kV

June 24, 2013

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0134. The Interconnection Customer proposes a 211.3 MW natural gas combined cycle facility interconnecting at the Podick 100 kV substation in the Ohio area. The request was received on June 06, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 111 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,642,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Lashaway to Podick 100kV Line is overloaded to 166.4% of its 127 MVA emergency rating for the loss of the Wolfe to Barrett 500kV. The project contributes 58.7% DFAX (123.9 MW) to this overload.

The Piercefield to Podick 100kV is overloaded to 162.3% of its 128 MVA emergency rating for the loss of the Lashaway to Podick 100kV. The project contributes 100.0% DFAX (211.3 MW) to this overload.

The Craigmont to Ltg Pleasant Hill 100kV Line is overloaded to 125.4% of its 150 MVA emergency rating for the loss of the Barrett to Norwich 500kV L. The project contributes 5.7% DFAX (12.1 MW) to this overload.

The Federal Hts to Ltg Pleasant Hill 100kV is overloaded to 118.3% of its 148 MVA emergency rating for the loss of the Barrett to Norwich 500kV. The project contributes 5.7% DFAX (12.1 MW) to this overload.

The Plant Gaston to Redfield 100kV is overloaded to 111.9% of its 130 MVA emergency rating for the loss of the Lashaway Substation to Podick 100kV. The project contributes 6.9% DFAX (14.5 MW) to this overload.

The Washington County Cogeneration Facility to Lashaway 100kV is overloaded to 111.2% of its 124 MVA emergency rating for the loss of the Smartsville to Stoney Ridge 100kV. The project contributes 49.4% DFAX (104.3 MW) to this overload.

The Archline to Plant Gaston 100kV is overloaded to 106.0% of its 123 MVA emergency rating for the loss of the Lashaway to Podick 100kV. The project contributes 6.9% DFAX (14.5 MW) to this overload.

For the loss of Lashaway to Podick 100kV, the Arcline to Charles City Rd 100kV loads to 104.3% of its 119 MVA rating. Project DFAX: 6.9%.

For the loss of Sour Lake to Firstenergy Mad River 100kV Circuit 2, the Firstenergy Mad Riv to Sour Lake 100kV loads to 103.5% of its 151 MVA rating. Project DFAX: 6.9%.

Sour Lake to Firstenergy Mad River Substation 100kV Line 2: 103.5% loading (151 MVA) following the outage of the Firstenergy Mad Riv to Sour Lake 100kV Ckt 1; DFAX = 6.9%.

For the loss of Barrett to Norwich Substation 500kV, the Larson St to Minco 230kV loads to 100.2% of its 683 MVA rating. Project DFAX: 20.9%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Podick to Lashaway 100kV	Rebuild line with higher-capacity conductor	\$4,342,000	100% (\$4,342,000)
Podick to Piercefield 100kV	Rebuild line with higher-capacity conductor	\$11,192,000	100% (\$11,192,000)
Craigmont to Ltg Pleasant Hill 100kV Line	Rebuild line with higher-capacity conductor	\$1,996,000	76% (\$1,509,000)
Federal Hts to Ltg Pleasant Hill 100kV	Rebuild line with higher-capacity conductor	\$1,479,000	76% (\$1,118,000)
Plant Gaston to Redfield 100kV line	Rebuild line with higher-capacity conductor	\$4,178,000	100% (\$4,178,000)
Washington County Cogeneration Facility Substation to Lashaway 100kV L	Rebuild line with higher-capacity conductor	\$1,705,000	100% (\$1,705,000)
Plant Gaston to Archline 100kV	Reconductor line	\$1,339,000	100% (\$1,339,000)
Charles City Road to Archline 100kV	Reconductor line	\$1,778,000	100% (\$1,778,000)
Firstenergy Mad Riv to Sour Lake 100kV	Reconductor line	\$1,098,000	100% (\$1,098,000)
Firstenergy Mad Riv to Sour Lake 100kV Ckt 2	Reconductor line	\$1,111,000	100% (\$1,111,000)
Minco to Larson Street 230kV	Reconductor line	\$3,833,000	7% (\$272,000)
	Total allocated to project		\$29,642,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.